

DEPARTMENT OF MECHANICAL ENGINEERING

MECHATRONICS & IOT 24MEC51

EXERCISE NO: 14

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MECHATRONICS & IoT

ASSIGNMENT REPORT – 14

TITLE:

Design and develop an MQTT-Based Remote HVAC Monitoring System using ThingSpeak Cloud. The system remotely monitors HVAC parameters and stores sensor data on cloud platforms for analysis and control.

Done By:

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Project Overview:

The MQTT-Based Remote HVAC Monitoring System is designed to remotely monitor important HVAC (Heating, Ventilation, and Air Conditioning) parameters using MQTT communication and ThingSpeak Cloud.

The system uses an ESP8266 NodeMCU as the main controller and a DHT11 sensor to measure temperature and humidity. The sensor data is processed by the ESP8266 and transmitted wirelessly to the cloud.

MQTT provides lightweight communication between the monitoring device and the cloud/IoT system. ThingSpeak stores the collected data and provides graphical visualization for analyzing HVAC environmental conditions.

The system helps users monitor HVAC conditions remotely and can be extended to automatically control HVAC equipment based on temperature and humidity.

Problem Statement

Traditional HVAC systems often require manual monitoring of temperature and humidity. Continuous monitoring becomes difficult when HVAC equipment is installed in large buildings, industries, server rooms, laboratories, or other remote locations.

The major problems include:

- Lack of continuous HVAC monitoring.
- Manual checking of temperature and humidity.
- Delayed detection of abnormal conditions.
- Difficulty in accessing HVAC data remotely.
- Lack of historical environmental data.
- Inefficient HVAC operation.
- Increased energy consumption.

Therefore, an IoT-based remote monitoring system is required to continuously collect HVAC parameters and store them on a cloud platform for monitoring and analysis.

Proposed Solution

The proposed system consists of:

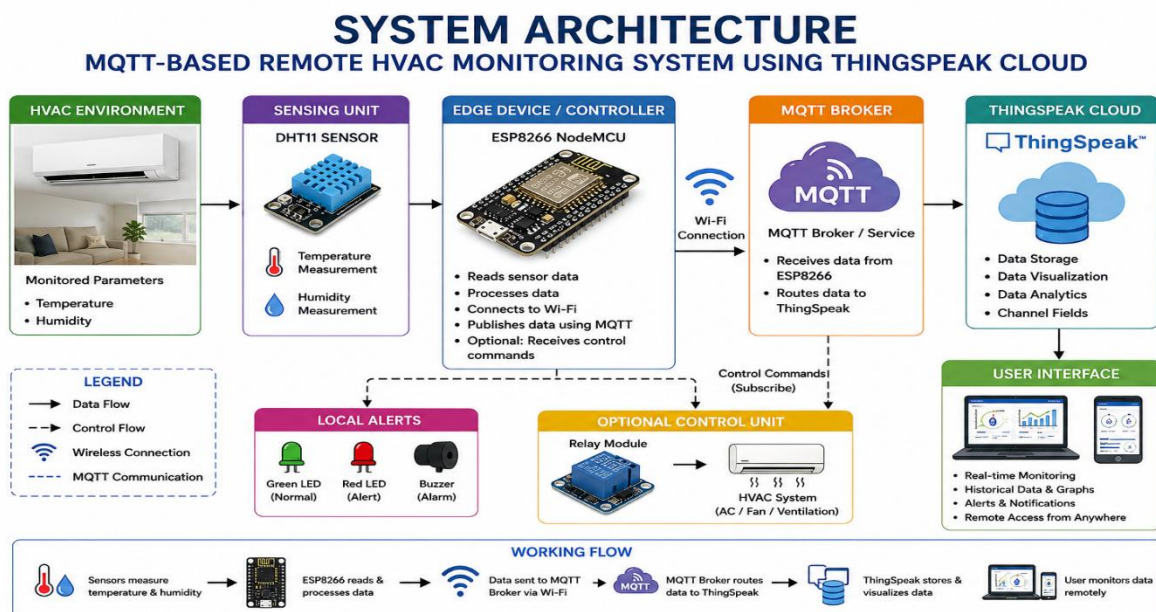
- ESP8266 NodeMCU
- DHT11 temperature and humidity sensor
- Wi-Fi connection
- MQTT communication
- ThingSpeak Cloud
- Optional relay/HVAC control unit

The DHT11 measures temperature and humidity and sends the readings to the ESP8266.

The ESP8266 processes the readings and connects to the Internet through Wi-Fi. The data is transmitted using the MQTT communication protocol and stored/visualized through the ThingSpeak cloud environment.

The user can access the data remotely through a computer or mobile device.

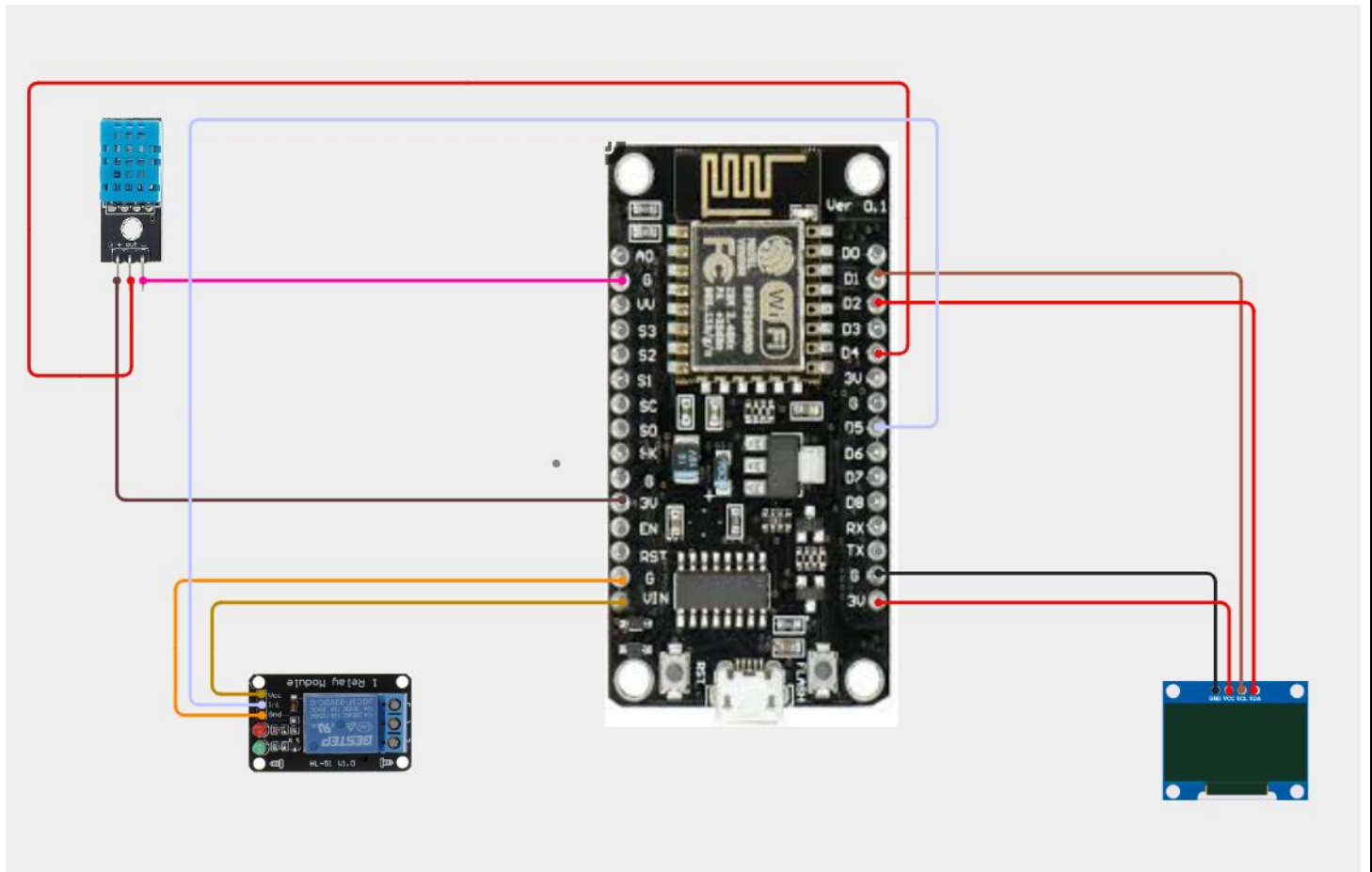
System Architecture



Circuit Table

S.No.	Component	Pin/Terminal	ESP8266 Connection
1	DHT11	VCC	3.3V
2	DHT11	GND	GND
3	DHT11	DATA	D4 (GPIO2)
4	Green LED	Anode	D1 through 220 Ω
5	Green LED	Cathode	GND
6	Red LED	Anode	D2 through 220 Ω
7	Red LED	Cathode	GND
8	Buzzer	Positive	D6 (GPIO12)
9	Buzzer	Negative	GND
10	Relay Module	IN	D5 (GPIO14)
11	Relay Module	GND	GND
12	Relay Module	VCC	Suitable module supply

Circuit Design



Algorithm

1. START
2. Initialize ESP8266 NodeMCU.
3. Initialize DHT11 sensor.
4. Initialize LEDs, buzzer and optional relay.
5. Connect ESP8266 to Wi-Fi.
6. Connect ESP8266 to MQTT broker/service.
7. Configure ThingSpeak channel and fields.
8. Read temperature from DHT11.
9. Read humidity from DHT11.
10. Check whether the sensor readings are valid.
11. Compare temperature and humidity with predefined thresholds.
12. If temperature and humidity are within the normal range:
 - Set status = NORMAL.
 - Turn ON Green LED.
 - Turn OFF Red LED.
 - Turn OFF Buzzer.
13. Else if readings are in the warning range:
 - Set status = WARNING.
 - Turn ON warning indication.
14. Else:
 - Set status = CRITICAL.
 - Turn ON Red LED.
 - Turn ON Buzzer.

15. Prepare the sensor data.
16. Publish the data using MQTT.
17. Send/store the data in ThingSpeak.
18. Display the readings on the cloud dashboard.
19. Wait for the next measurement interval.
20. Repeat the monitoring process continuously.
21. STOP.

Coding

```
#include <ESP8266WiFi.h>
```

```
#include <DHT.h>
```

```
#include <ThingSpeak.h>
```

```
// --- Configuration Credentials ---
```

```
const char* ssid = "Harish Kumar K K";
```

```
const char* password = "Mavboiii@10"; // Add your WiFi password
```

```
unsigned long myChannelNumber = 3459318;    // Add your ThingSpeak  
Channel ID (Number only)
```

```
const char* myWriteAPIKey = "OAQCYUA6UGNIY18Y"; // Add your  
ThingSpeak Write API Key
```



```
// --- Sensor Configuration ---
```

```
#define DHTPIN D5      // Pin D4 (GPIO2) connected to DHT sensor data  
pin
```

```
#define DHTTYPE DHT11  // Change to DHT22 if using a DHT22 sensor
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
WiFiClient client;
```

```
// --- Timing Settings ---
```

```
unsigned long lastTime = 0;
```

```
const unsigned long timerDelay = 20000; // ThingSpeak free tier requires a  
minimum 15s interval
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    dht.begin();
```

```
    // Connect to Wi-Fi
```

```
    WiFi.begin(ssid, password);
```

```
    Serial.print("Connecting to WiFi");
```

```
    while (WiFi.status() != WL_CONNECTED) {
```

```
    delay(500);

    Serial.print(".");

}

Serial.println("\nConnected to WiFi!");


// Initialize ThingSpeak

ThingSpeak.begin(client);

}


void loop() {

    if ((millis() - lastTime) > timerDelay) {

        // Read relative humidity (%) and temperature (°C)

        float humidity = dht.readHumidity();

        float temperature = dht.readTemperature();


        // Verify reading validity

        if (isnan(humidity) || isnan(temperature)) {

            Serial.println("Failed to read from DHT sensor!");

        } else {

            Serial.print("Temperature: ");

            Serial.print(temperature);
```

```
Serial.print(" °C | Humidity: ");

Serial.print(humidity);

Serial.println(" %");

// Set values to ThingSpeak fields

ThingSpeak.setField(1, temperature);

ThingSpeak.setField(2, humidity);


// Write data to ThingSpeak channel

int responseCode = ThingSpeak.writeFields(myChannelNumber,
myWriteAPIKey);

if (responseCode == 200) {

    Serial.println("ThingSpeak update successful.");

} else {

    Serial.print("Error updating ThingSpeak. HTTP error code: ");

    Serial.println(responseCode);

}

}

lastTime = millis();

}

}
```

System Working

The DHT11 sensor is placed in the HVAC environment to continuously measure temperature and humidity.

The ESP8266 reads the sensor values and processes them.

The controller then connects to the Wi-Fi network and communicates with the MQTT service.

Normal Condition

When the temperature and humidity are within the defined limits:

- Green LED → ON
- Buzzer → OFF
- Status → NORMAL

Warning Condition

When the environmental values approach an abnormal range:

- Warning indication → ON
- Status → WARNING

Critical Condition

When the temperature or humidity exceeds the critical threshold:

- Red LED → ON
- Buzzer → ON
- Status → CRITICAL

The sensor values are published through MQTT and made available to the cloud monitoring system.

ThingSpeak stores the readings and generates graphs for analysis.

Bills Of Material

S.No.	Component	Specification	Quantity	Purpose
1	ESP8266 NodeMCU	Wi-Fi Microcontroller	1	Main Controller
2	DHT11 Sensor	Temperature & Humidity	1	HVAC Parameter Monitoring
3	Green LED	5 mm	1	Normal Indication
4	Red LED	5 mm	1	Critical Indication
5	Resistor	220 Ω	2	LED Current Limiting
6	Buzzer	3.3/5 V	1	Alarm
7	Breadboard	Standard	1	Prototype Assembly
8	Jumper Wires	Male-Male	1 Set	Circuit Connections
9	USB Cable	Micro USB	1	Programming & Power
10	Wi-Fi Router/Hotspot	2.4 GHz	1	Internet Connectivity

Prototype Implementation

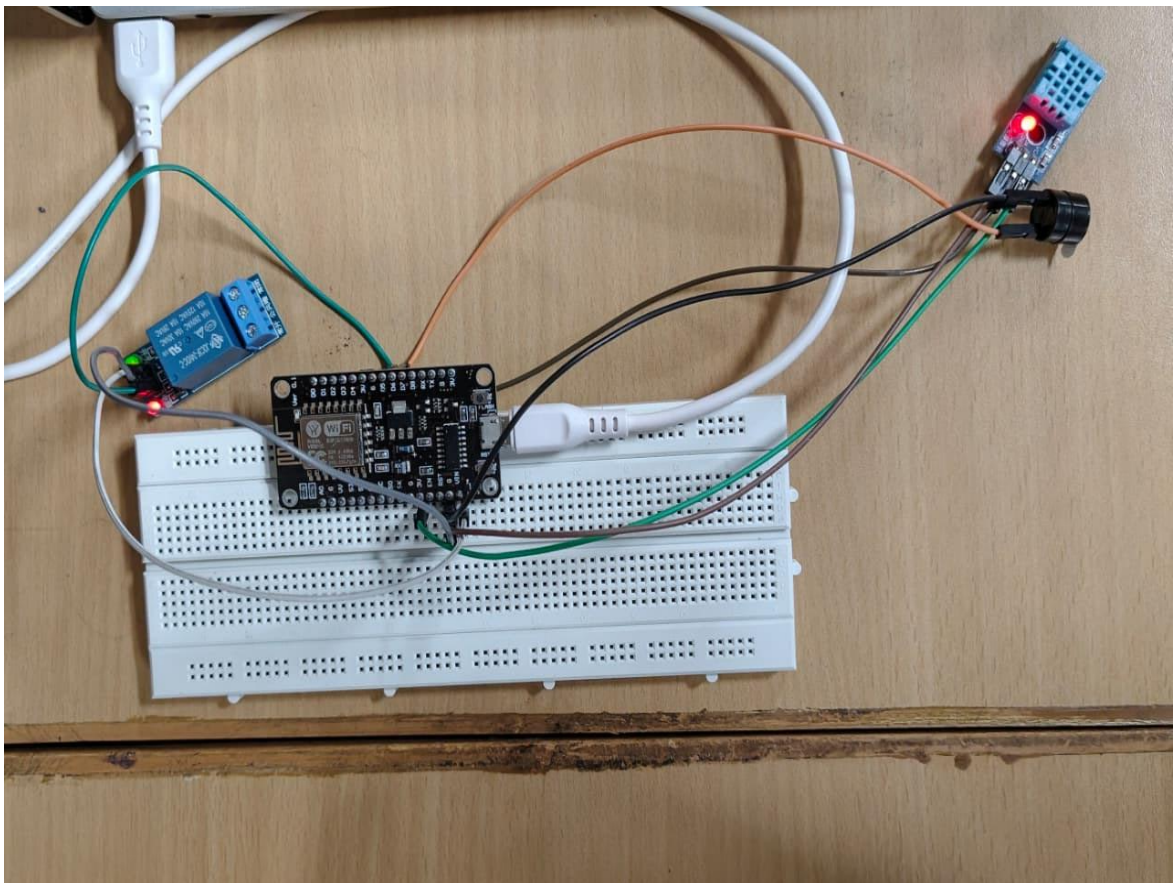
The prototype is assembled using an ESP8266 NodeMCU, DHT11 sensor, LEDs, buzzer, breadboard, and jumper wires.

The DHT11 sensor is positioned inside the area being monitored. It continuously measures temperature and humidity.

The ESP8266 reads these values and evaluates the HVAC environmental condition. It then connects to the Internet through Wi-Fi.

Sensor readings are transmitted using MQTT and stored/displayed through ThingSpeak Cloud.

The user can access the ThingSpeak dashboard remotely using a PC or mobile device.



Results & Performance Analysis

The system successfully demonstrates remote monitoring of HVAC environmental conditions.

Parameter	Result
Temperature Monitoring	Yes
Humidity Monitoring	Yes
Wi-Fi Connectivity	Yes
MQTT Communication	Supported
Cloud Storage	ThingSpeak
Remote Monitoring	Yes
Graphical Data Analysis	Yes
Alert System	Yes
Historical Data	Supported

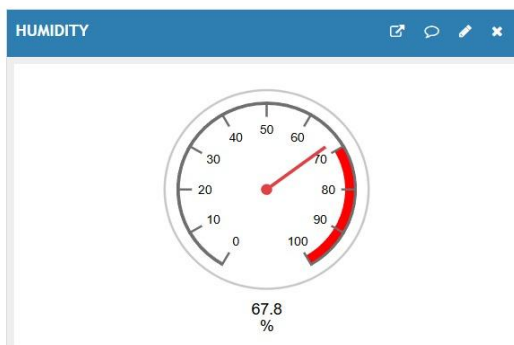
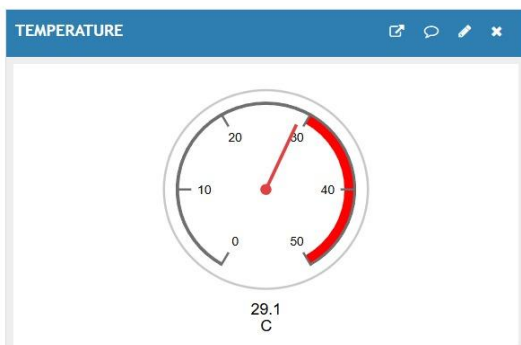
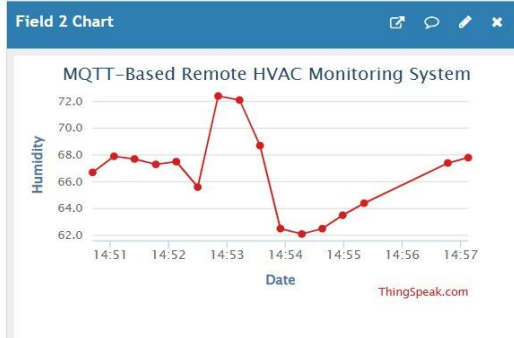
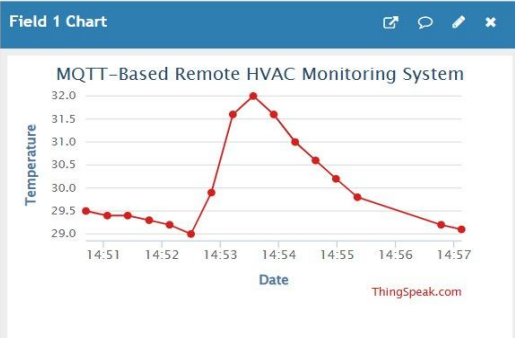
Performance Analysis

The system allows HVAC parameters to be monitored without physically visiting the HVAC location.

MQTT provides lightweight communication suitable for IoT applications, while ThingSpeak provides cloud-based storage and visualization.

Historical graphs can help identify temperature and humidity patterns and provide useful information for HVAC optimization.

```
Humidity      : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.30 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.30 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.30 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.40 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.40 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.40 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!  
-----  
Temperature : 27.40 C  
Humidity    : 74.30 %  
HVAC Status : COOL  
Data Sent to ThingSpeak  
Connecting to ThingSpeak MQTT... Connected!
```



MQTT-Based Remote HVAC Monitoring System

Channel ID: 3459318
Author: mwa0000037545071
Access: Private

[Private View](#) [Public View](#) [Channel Settings](#) [Sharing](#) [API Keys](#) [Data Import / Export](#)

[+ Add Visualizations](#)

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Channel 2 of 2 < >

Channel Stats

Created: about an hour ago
Last entry: less than a minute ago
Entries: 11

